

# The Ephemeris

July 2014

Volume 25 Number 07 - The Official Publication of the San Jose Astronomical Association



## In Memory of Robert Armstrong

SJAA Readers and Friends,



Dr. Robert Armstrong passed away courageously and peacefully on Tuesday, 17 June, after suffering an accidental fall at his home in Los Gatos in late May. Dr. Armstrong served as a member of the board of directors and treasurer of the SJAA for many years. He was known as a humble gentleman and consummate professional who was respected by all.

Robert had gone observing at Fremont Peak just days before the accident, and had been proud of his association with the SJAA and the local amateur astronomical community. Over the years, Robert participated not only with the running of the SJAA, but also with outreach to the public. When the club began participating in local community festivals, Robert was one of the first to sign up, even after staying out late the previous night, observing until the early morning hours. He was a major supporter of the club acquiring a high quality solar telescope and starting the Solar Program. He is the most recent recipient (2012) of the SJAA's highest symbol of recognition, the Dr. A. B. Gregory Award.

The SJAA, the local community and all SJAA members past and present have lost a true friend, and we are truly saddened by his passing.

Respectfully,  
Rob Jaworski

## July 2014 Events

### Sunday, July 06

Solar observing: 2-4PM

Fix-It Day: 2-4PM

### Saturday, July 12

Board of Directors Meeting: 6-7:30pm

General Meeting: 7:30-10pm

### Friday, July 18

Beginner Astronomy Class: 8:30-9:30pm

In-Town Star party (Houge): 9:30-11:30pm

Binocular Stargazing (RCDO): 8:45-10:45pm

### Saturday, July 19

Starry Nights Star Party: 9:30-11:30pm

Ranch Cañada del Oro

### Saturday, July 26

Henry Coe—Dark-Sky Weekend

### Friday, August 01

In-Town Star party (Houge): 9:15-11:15pm

### Sunday, August 03

Solar observing: 2-4PM

Fix-It Day: 2-4PM

*SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.*

## Deep Sky Scavenger Hunt

Mendoza Ranch June 28, 2014



*Group Photo of the DSSH Participants. See article on Page 3*

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# Rancho Canada del Oro (RCDO)

## Starry Nights Star Party June 21, 2014

From Tom Pillar

Rancho Canada del Oro is a Santa Clara County Open Space Authority (OSA) Open Space Preserve and is located just south of San Jose but North of Morgan Hill. This site offers amazing views of the Milky Way free of the light pollution that is present in the city. In conjunction with SJAA, OSA hosts monthly Star Parties at RCDO and the event has been gaining in popularity every month for the last year.

The June 21st evening sky was pretty much clear as the sun dropped below the horizon and twilight commenced. Dark, clear, night skies followed. SJAA members set up approximately 15 scopes / binoculars and were later joined by more than 70 visitors.

Saturn and Mars were first up and as the night progressed SJAA members turned their scopes on some of the brighter stars; Mizar, Alberio, Vega, Arcturus, Spica and Antares. At 9:51pm we were treated to an International Space Station (ISS) flyby; it stayed visible for quite a long time and was a real crowd pleaser. By 10:30pm it was dark enough to see many of the higher magnitude deep sky objects; I was able to observe M13, M4, M8, M22, M20, M17 and the asterism, Collinder 399 (Coathanger) to name a few. SJAA's Teruo Utsumi had asteroids Vesta and Ceres located and helped me zero them in on my scope; a first for me to see an asteroid.

Around 11:30pm the Milky Way became very visible in the southern sky and it was nice to just sit back and take it all in before packing up for the night.

I consider myself lucky to be able to visit a dark sky site like RCDO so close to home; it is less than a 1 hour drive from just about anywhere in South Bay. I am already looking forward to next month.

For more information about Open Spaces:

Check out Open Space Authority

<http://www.openspaceauthority.org/>

and Rancho Canada del Oro <http://www.openspaceauthority.org/trails/rancho.html>



Above: R-L; Teri Rogoway (OSA Program Coordinator), and OSA volunteers Victor Rising and Joan Murphy



06.21.2014



06.21.2014

# Deep Sky Scavenger Hunt (DSSH)

Mendoza Ranch June 28, 2014

From Tom Piller



Above L-R: SJAA Board Members Rob Jaworski, Dave Ittner, Ed Wong, Teruo Utsumi



The SJAA's Deep Sky Scavenger Hunt (DSSH) was brought about through the tireless efforts of Ed Wong over the last several months. The concept was to promote a better understanding of Deep Sky Objects in our night sky and to prep participants for the summer viewing that lies ahead. Both of these objectives were accomplished through the well thought out lists of DSOs that Ed provided and the viewing event that followed on June 28th..

There were three lists; Beginner, Intermediate and Advanced where Ed had sorted the objects, for the most part, in viewing order based on when the objects would set in the West and rise in the East. This really helped organize the viewing to the benefit of all participants.

The Mendoza Ranch DSSH event encompassed two evenings; the first of which was a trial run with coaching by Ed and Dave Ittner and the second evening was the actual scavenger hunt. The trial run on June 27th was plagued by what seemed like hurricane force winds but the group of 14 was pretty much able to get familiar with the objects on the lists of DSOs they had selected.

The main event on June 28th started around 9pm, lasted until midnight and was attended by more than 24 participants using 16 telescopes. There was very little wind, low moisture and clear skies; a perfect viewing night. Lots of talking, strategizing and laughing throughout the night. The whole event was tons of fun and I personally had a blast.

**Well Done Ed!**

**And the winners were:** (winners of 1st, 2nd and 3rd place were determined by those who found all objects or highest count of each the 35 object lists by midnight)

**Beginning List** (all objects found)

Winner - Ido Greiman  
Runner up - Guy Ronen

**Intermediate List** (highest count)

Winner - Gary Chock  
Runner Up - (Tied) Jose Marte, Gene McCreary

**Advanced List** (highest count)

Winner - Alex Kramarov  
Runner Up - Ryan Lu

## Ceres and Vesta

From Teruo Utsumi

S&T and other sources have noted that the asteroids Ceres and Vesta are now visible. They'll both continue to be in the same binocular view for a while. I've seen them the past couple weeks and, courtesy of Tom Piller, saw them at Rancho through his telescope in a 2° field of view. It turns out they'll be closest together around July 4th. Because they're small, you have to know exactly where they are. Otherwise there's no way to distinguish them from stars.

If you haven't seen them yet, you should! Being able to see both Ceres and Vesta in the same view is a rare occurrence. Although they're both in the asteroid belt, their orbits are somewhat different. It turns out they get close to each other only once every 16 years. The next time they'll be within 5° of each other will be Feb. 2030 and, drum roll please, the next time they'll be within 2° is 2081!

So how do I know that? "Check out the continuation of this article in the SJAA Blog My Favorite Things".

# PINNACLES EXPLORATORY TRIP May 31, 2014

From Ed Wong

This a report on an exploratory trip to the Pinnacles wherein nine (9) SJAA members (Ed W, Dave I, Tom P, Gary C, Guna P, Lee H, Teruo U, Manoj K, Bharath K,) gathered at Pinnacles National Park to check out the dark sky viewing.

The campsites we reserved were on opposite sides of the access road. Our best campsite was large, flat and big enough to fit all of us so we decided to setup all tents and gear on that site and park overflow cars at the opposite site to stay within the 5 car parking limit.

As we were setting up it became obvious to the other campers that we were not going to be a normal group. Needless to say the big Dobs and Big Binoculars got the attention of those around us which gave us a good opportunity to talk about the club and invite people to our website and Houge Park. Of course the other campers wanted to have some views through our scopes too.

Following is a list of OTAs SJAA members brought to the campsite:

- ◇ Dave I – 17.5 inch Dobstuff Dob
- ◇ Lee H – 25x150mm battle ship binoculars
- ◇ Ed W- Oberwerk 28x110mm binoculars
- ◇ Tom P – Televue NP101 & Astrotech AT72 refractors
- ◇ Gary C- Astrotech AT72 refractor & Celestron 20x80 binoculars
- ◇ Teruo U – 18 inch Obsession Dob
- ◇ Manoj – 16 inch Hubble Optics Dob & 8inch? Dobstuff Dob
- ◇ Guna P – Explore Scientific ED102
- ◇ Bharath K – Orion ED80
- ◇ Nhan N – Orion ED80

The night started off with viewing the Moon and Jupiter. I spotted Jupiter with my binos just before sunset and Dave was close be-



hind. We were able to see the bands of Jupiter and the sun was still up!!! The crowd gathered around to see views offered of the Moon, Mars, Jupiter and Saturn just as the sun set. A little later in the evening, as a bonus, we were able to see the Moon occult the bright star SAO 96786. The occult took place just minutes before the moon set. Lucky! By 11pm the crowd started to dwindle down and I was able to get my first SQM reading of 20.56 although there were still many lights glaring from the nearby campers and the restrooms (by comparison RCDO is approximately 20.50). Around 1am the campers had gone to bed, the lights were out and I took a second SQM reading of 21.77!!! The Milky Way was blazing across the sky.

If you have never had a chance to view the Milky Way under dark skies it is truly a sight to behold. With my binos I panned across the M8 – Lagoon Nebula, over to M24, through the Sagittarius Star cloud and then up to the M11 – Wild Duck cluster. It is amazing how many stars are visible even with the dark dust lanes in between them. Other nice targets viewed during the evening were: the Markarian Chain, M83, M104, The Leo Triplet, M31, M51, M64, M13, M92, M95, M96, M81, M82, M97, M105, M106, M108, M109. Bonus objects were the NG7000 – North American Nebula, Eastern & Western Veil Nebulas and all looked fantastic in the binos.

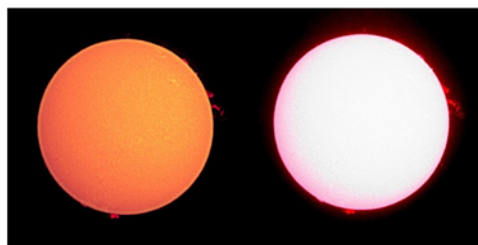
By 2:30am it was getting cold and most of us were starting to pack up or crawl into sleeping bags for the night.

Parting thoughts about the trip are that overall it was a great, dark sky, viewing experience. We learned things about the site that will help us plan for future trips and we even got a chance to introduce the other campers to SJAA.

## Solar Observing June 1st 2014

From Michael Packer

### Holy Solar H-Flares!



*It was a warm day under the sun but the H-Alpha Solar flares did not disappoint. To the left are two exposures taken with a LX5 point and shoot held at the eyepiece.*

### Observe The Sun Safely!

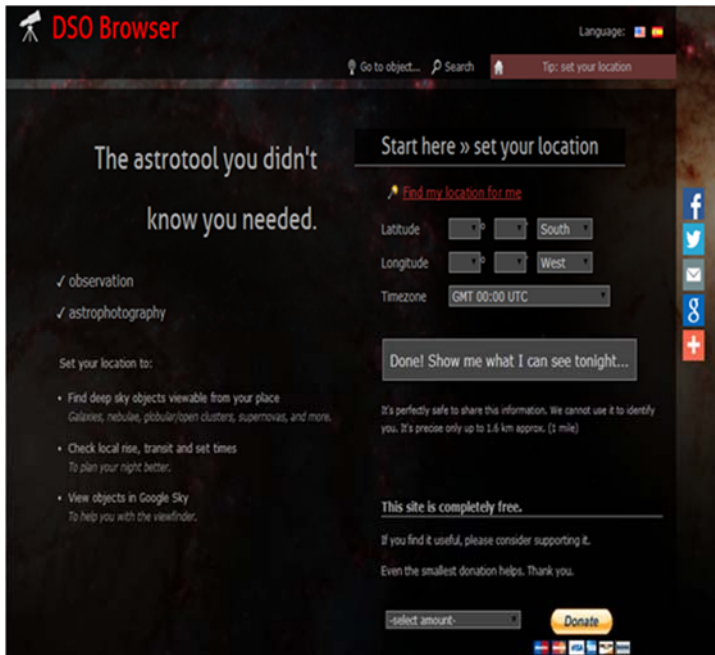
**Never look at the Sun without a proper filter!**

Solar Programs are held 1st Sunday of every Month 2:00-4:00PM at Houge Park weather permitting.



# Deep Sky Objects Browser (DSO)

From Ed Wong and Tom Piller



## Screenshot from DSO browser

*Editor's Note: As an example of how this tool might be used, in preparation for the Half Messier Marathon, Ed input the information that was related to Mendoza Ranch including date, time and horizon. The browser generated a printable list of the Messier Objects that would be viewable from Mendoza Ranch on 3/29/2014 between 8pm-12am and 20 degrees above the horizon.*

New to astronomy? Just bought a new telescope and trying to learn what is in the night sky to look at? Been viewing for a while and want to find some different objects to look at? Trying to plan a observing session for a night and want a list of the objects that will be viewable from your location? If you answered yes to any of these questions there is a new tool that can be of real help.

The tool is called "Deep Sky Objects Browser" and it's free. Go to <http://dso-browser.com/>

This is a web based tool that uses the large object database from the Saguaro Astronomy Club Database Ver. 8.1. The browser lets you create filters in order to generate a list of objects that will be viewable in the night sky according to the filters you use. The filters include: viewing time/location/date, minimum altitude, apparent magnitude, apparent size, surface brightness, object types, object coordinates and catalogue (Messier, NGC, IC, Caldwell, etc). It also sorted which objects would be setting in the west first; very cool.

DSO is a tool that can be used to organize viewing sessions and make our viewing opportunities for objects of the night sky just a little more efficient. Check it out and you just might find you have also learned something new about the night sky.

DSO Creator's Background: **Sebastián García Rojas**, grew up and currently resides in Buenos Aires, Argentina. He attended the **Universidad Nacional de Buenos Aires (UBA)** and his occupation is software programming, mainly oriented

to web applications.

*Editor's Note: After reading Ed's article last March I corresponded with Sebastian Rojas (DSO Creator) via email and he was kind enough to take the time to answer my questions.*

Piller: Why did you create the DSO web page?

SGR response: "At that time, I was a freelance programmer and had plenty of time for my favorite hobbies: astronomy and photography. But there just wasn't any good free tool to help me plan my observation/astrophotography session, so I went ahead and just did it myself. After a while, I polished it up and made it public so all other amateur astronomers could benefit from it."

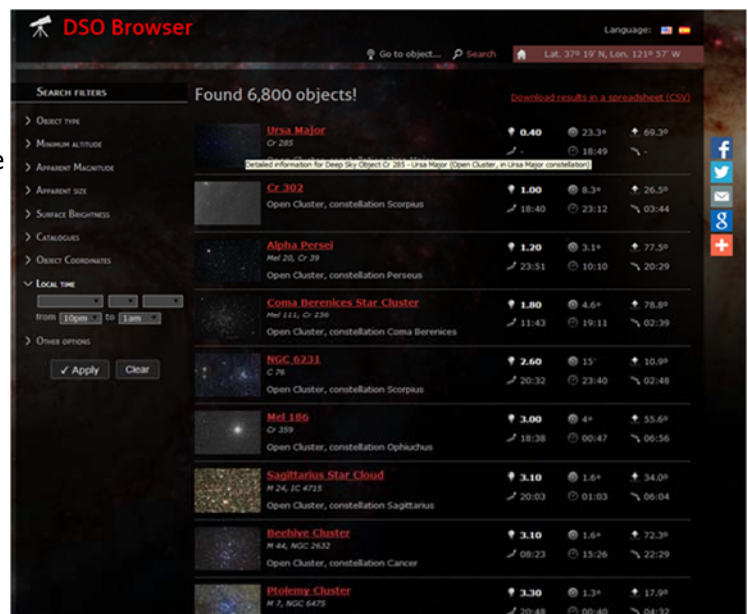
Piller: Anything else you think might be of interest to the SJAA Ephemeris readers?

SGR response: "The basics of the DSO Browser were quite simple to do, but for the first time in my life I found myself using advanced mathematics to do some complex calculations in real life :). I have plenty of ideas for the site and I would really encourage people to write to me through the contact page (<http://dso-browser.com/contact>) to share their thoughts - recommendations, suggestions, anything is welcome.

Another hobby of mine is photography and I'd really like to have more free time to be able to do some serious astrophotography again.

I'm currently working on another project that's not quite similar to the DSO Browser but could be of interest to you too: <http://finddealsfor.me>. It's basically a search engine to help users find good products with free worldwide shipping around the world. It's a decent technical challenge as well: we need to be able to handle millions of entries efficiently to have the most up to date database possible.

Both the DSO Browser and FindDealsFor.me are projects I work on in my free time."



# KID SPOT



## Kid Spot Jokes:

- ◆ **When is the moon the heaviest?**  
(When it's full!)
- ◆ **How does the man-in-the-moon cut his hair?**  
(Eclipse it!)

## Kid Spot Quiz:

1. **What is the name of the first space probe to leave the solar system?**
2. **What is the month and year did this space probe leave the solar system?**



Photo Credit: Wikipedia

## Kid Spot Night Sky Challenge: July 2014

See if you can spot the following objects in the sky:

- ⇒ Saturn – All month
- ⇒ Mars – Will be very bright!
- ⇒ Antares - Star in the constellation Scorpio
- ⇒ M4—Globular cluster close to Antares in constellation Scorpio

<http://skyandtelescope.com/observing/atalance>

## Constellations

### Scorpius

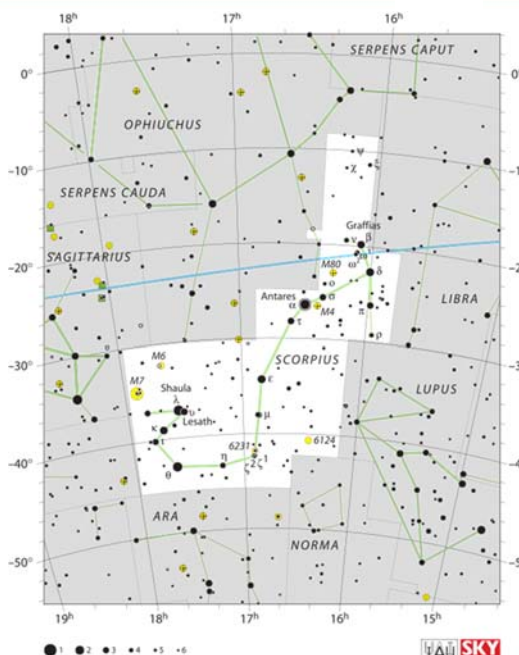
Sometimes known as Scorpio, Scorpius is one of the constellations of the zodiac. Its name is Latin for scorpion, and its

symbol is  (Unicode ♏). It lies between Libra to the west and Sagittarius to the east. It is a large constellation located in the southern hemisphere near the center of the Milky Way.

Scorpius contains many bright stars, including Antares, Graffias, a triple star; δ Dschubba, the forehead; and marking the tip of the scorpion's curved tail are λ Shaula and υ Lesath, whose names both mean "sting." Given their proximity to one another, λ and υ are sometimes referred to as the Cat's Eyes.

Deep-sky objects - Due to its location on the Milky Way, this constellation contains many deep-sky objects such as the open clusters Messier 6 (the Butterfly Cluster) and Messier 7 (the Ptolemy Cluster), NGC 6231 and the globular clusters Messier 4 and Messier 80.

Messier 80 (NGC 6093) is a globular cluster of magnitude 7.3, 33,000 light-years from Earth. It is a compact Shapley class II cluster; the classification indicates that it is highly concentrated



and dense at its nucleus. M80 was discovered in 1781 by Charles Messier.

In Greek Mythology the myths associated with Scorpio almost invariably also contain a reference to Orion. According to one of these myths it is written that Orion boasted to goddess Artemis and her mother, Leto, that he would kill every animal on the earth. Although Artemis was known to be a hunter herself she offered protection to all creatures. Artemis and her mother Leto sent a scorpion to deal with Orion. The pair battled and the scorpion killed Orion. However, the contest was apparently a lively one that caught the attention of the king of the gods Zeus, who later raised the scorpion to heaven and afterwards, at the request of Artemis, did the same for Orion to serve as a reminder for mortals to curb their excessive pride. So every winter

Orion hunts in the sky, but every summer he flees as the constellation of the scorpion comes.

Source: Wikipedia

### Kid Spot Quiz Answers:

- 1) **Voyager 1**
- 2) **September 2013**

# Club Updates

## SJAA Library

From Sukhada Palav

Thanks to a generous donation from one of our board members, Ed Wong...SJAA Library now has 14 awesome books for the little ones! (listed below).

1. Find the Constellations
2. Zoo in the Sky: A Book of Animal Constellations
3. The Everything Kids' Astronomy Book
4. Glow-in-the-Dark Constellations
5. The Kids Book of the Night Sky
6. Exploring the Night Sky: The Equinox Astronomy Guide for Beginners
7. There's No Place Like Space: All About Our Solar System
8. Once Upon a Starry Night: A Book of Constellations
9. Telescope Power: Fantastic Activities & Easy Projects for Young Astronomers
10. The Moon Book
11. National Geographic Kids First Big Book of Space
12. National Geographic Readers: Planets
13. Faraway Worlds
14. Night Sky Atlas

All of these books are very kid-friendly, with lots of fun pictures and easy explanations. Stop by the book cabinet to check these out, next time you are Houge Park for one of our events! I hope our budding astronomers group will enjoy reading these books!!!

For more information on SJAA Library, please check out - <http://www.sjaa.net/sjaa-library/>

If you have any questions or comments, I can be reached at:

[librarian.sjaa@gmail.com](mailto:librarian.sjaa@gmail.com).

## Advanced Loaner Telescope & QuickSTART Programs

From Dave Ittner

<http://www.sjaa.net/advanced-loaner-telescope-program/>

I am glad to report that the storage room is nearly empty! With 7 of the 8 Quick Start scopes out and a lot of the Advanced Loaner Program items out on loan as well, the storage room looks empty.

I do have some bad news to report. The club's 20" scope suffered a significant set back when the bottom of the mirror box gave way. The mirror, mirror cell, and metal bottom dropped about an inch or so. This caused a few extra chips in the mirror as well. Chris K. is looking into making repairs.

## Fix-It Program

From Ed Wong

The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at. It can be any type of issue – such as broken scopes whose owners need advice, help with collimation, or even a quick session on how to use it.

The session is held every first Sunday of the month, from 2 to 4 PM (coinciding with the Solar Observing sessions) at Houge Park.

The June 1st Fixit Session was busy. Thanks to Phil, Tom Tanquary, Teruo and Dave for helping out. We had at least 3 people who came in 2 of them were non-members. Some needed their RC telescopes collimated, Phil graciously took care of them. Others had questions about their scope setups which Teruo, Tom and I took care of. Dave was busy checking in scopes that were brought back from loan.

## From the Board of Directors

### General Notices

One Board seat remains open. Contact Lee Hoglan [or any Board member] if interested.

### Announcements

#### SJAA 60 Years Old

SJAA will be 60 years old this November! The first meeting was held on December 6, 1954, at Herbert Hoover Junior High School. There might be a celebration brewing; keep an eye out for future announcements.

## Board Meeting Excerpts June 14, 2014

### In attendance

Rob Jaworski, Lee Hoglan, Ed Wong, Dave Ittner, Teruo Utsumi

### Pinnacles

Per Ed Wong the exploratory trip went well and there is a much better understanding of site conditions, availability and logistics for a future SJAA event at possibly the east end of the campground.

### Astronomical League (AL)

Some Board members expressed a desire for SJAA to join the Astronomical League. Rob Jaworski to take the lead on making contact and starting the discussion on how best to work with AL.

## Mobile Observatory Project

Teruo was contacted by the Mobile Observatory Project (funded by Kickstarter) for a possible collaboration in San Jose February 2015. Teruo will reconnect and inquire about their program and goals.

## Advanced Loaner Program Upgrade

Dave Ittner expressed a need to upgrade the loaner equipment program with an aim to more closely align future SJAA purchases with the equipment SJAA members are looking for. Ken Presti will take the lead on revisiting a member survey Harsh compiled in 2013.

## SJAA Contacts

|                        |                  |
|------------------------|------------------|
| President:             | Rob Jaworski     |
| Vice President:        | Lee Hoglan       |
| Treasurer:             | Michael Packer   |
| Secretary:             | Teruo Utsumi     |
| Director:              | Greg Claytor     |
| Director:              | Dave Ittner      |
| Director:              | Ed Wong          |
| Director:              | Rich Neuschaefer |
| Director:              | pending          |
| Beginner Class:        | pending          |
| Fix-it Program:        | Ed Wong          |
| Imaging SIG:           | Harsh Kaushikkar |
| Library:               | Sukhada Palav    |
| Loaner Program:        | Dave Ittner      |
| Ephemeris Newsletter - |                  |
| Editor                 | Sandy Mohan      |
| Production             | Tom Piller       |
| Publicity:             | Rob Jaworski     |
| Questions:             | Lee Hoglan       |
| Quick START            | Dave Ittner      |
| Solar:                 | Michael Packer   |
| School Events:         | Jim Van Nuland   |
| Speakers:              | Teruo Utsumi     |

E-mails: <http://www.sjaa.net/contact.shtml>

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Articles for publication should be submitted by the 20th of the previous month.

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## San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New**   ☐ **Renewal** (Name only if no corrections)

☐ **I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at:**

### Membership Type:

☐ Regular — \$20

☐ Regular with Sky & Telescope — \$53

☐ Junior (under 18) — \$10

☐ Junior with Sky & Telescope — \$43

<http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to  
**[sjaamemberships@gmail.com](mailto:sjaamemberships@gmail.com)**

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at <http://www.sjaa.net/join-the-sjaa/>

**Name:** \_\_\_\_\_

**Address:** \_\_\_\_\_

**City/ST/Zip:** \_\_\_\_\_

**Phone:** \_\_\_\_\_

**E-mail address:** \_\_\_\_\_